EXPERIMENT – 2

Encryption – Crypto 101 using TryHackMe Platform

Aim:
To understand the fundamentals of cryptography by performing hands-on exercises in the Crypto 101 room on the TryHackMe platform, including basic encryption, encoding, hashing, and cryptographic concepts.

Tools / Platform Required:
- Web browser (Chrome / Firefox)
- Internet connection
- TryHackMe account (free)
- Built-in TryHackMe terminal (AttackBox) or local Linux machine

Procedure:
Step 1: Login to TryHackMe
1. Open browser and go to: https://tryhackme.com
2. Login using your registered credentials.

Step 2: Open Crypto 101 Room
1. In the search bar, type Crypto 101
2. Click on the room named Crypto 101
3. Click Join Room

Step 3: Start the Machine (if required)
1. Click Start Machine / Start AttackBox
2. Wait for the virtual environment to load.

Step 4: Perform Cryptography Tasks
Complete the tasks provided in the room, such as:

4.1 Encoding & Decoding

(a) Base64 Encoding / Decoding
Purpose:
Base64 is an **encoding technique** used to represent binary data in readable ASCII format.
How to Solve
To Decode Base64
1. Copy the given Base64 string from the question.
2. Open the TryHackMe terminal.
3. Use the command:
   echo "Q3J5cHRvZ3JhcGh5" | base64 -d
4. The output will be the decoded plain text.

5. Enter the decoded text into the answer box.

To Encode Base64
echo "cryptography" | base64
Expected Output Example: Cryptography

(b) Hexadecimal Conversion
Purpose:
Hexadecimal is a base-16 encoding system commonly used in computing**.**
How to Solve
Hex to Text
echo 48656c6c6f | xxd -r -p

Text to Hex
echo "Hello" | xxd -p
Expected Output: Hello

4.2 Classical Ciphers
(a) Caesar Cipher
Purpose:
Caesar cipher shifts letters by a fixed number.
How to Solve
1. Identify the shift value (given or by trial).
2. Use online Caesar decoder or terminal tool.

Using terminal
echo "KHOOR" | tr 'A-Z' 'X-ZA-W'
(Shift = 3)
Expected Output: HELLO

(b) Shift-Based Encryption
Purpose:
Similar to Caesar cipher but may use different shift values.
How to Solve
1. Try different shift values until meaningful text appears.
2. Online tools can be used if allowed.

Example:
- Shift 1 → incorrect
- Shift 3 → readable text → correct answer

4.3 Hashing

(a) Understanding Hash Algorithms

Hash Type	Length
MD5	32 characters
SHA-1	40 characters
SHA-256	64 characters

(b) Identifying Hashes

How to Solve

1. Count the number of characters in the given hash.
2. Match it with the table above.

Example:
5d41402abc4b2a76b9719d911017c592

→ 32 characters → MD5

Generating Hashes

```
echo -n "hello" | md5sum
echo -n "hello" | sha1sum
echo -n "hello" | sha256sum
```

4.4 Encryption Concepts

(a) Encryption vs Hashing

Encryption	Hashing
Reversible	Irreversible
Uses keys	No keys
Used for secure communication	Used for password storage

(b) Symmetric vs Asymmetric Encryption

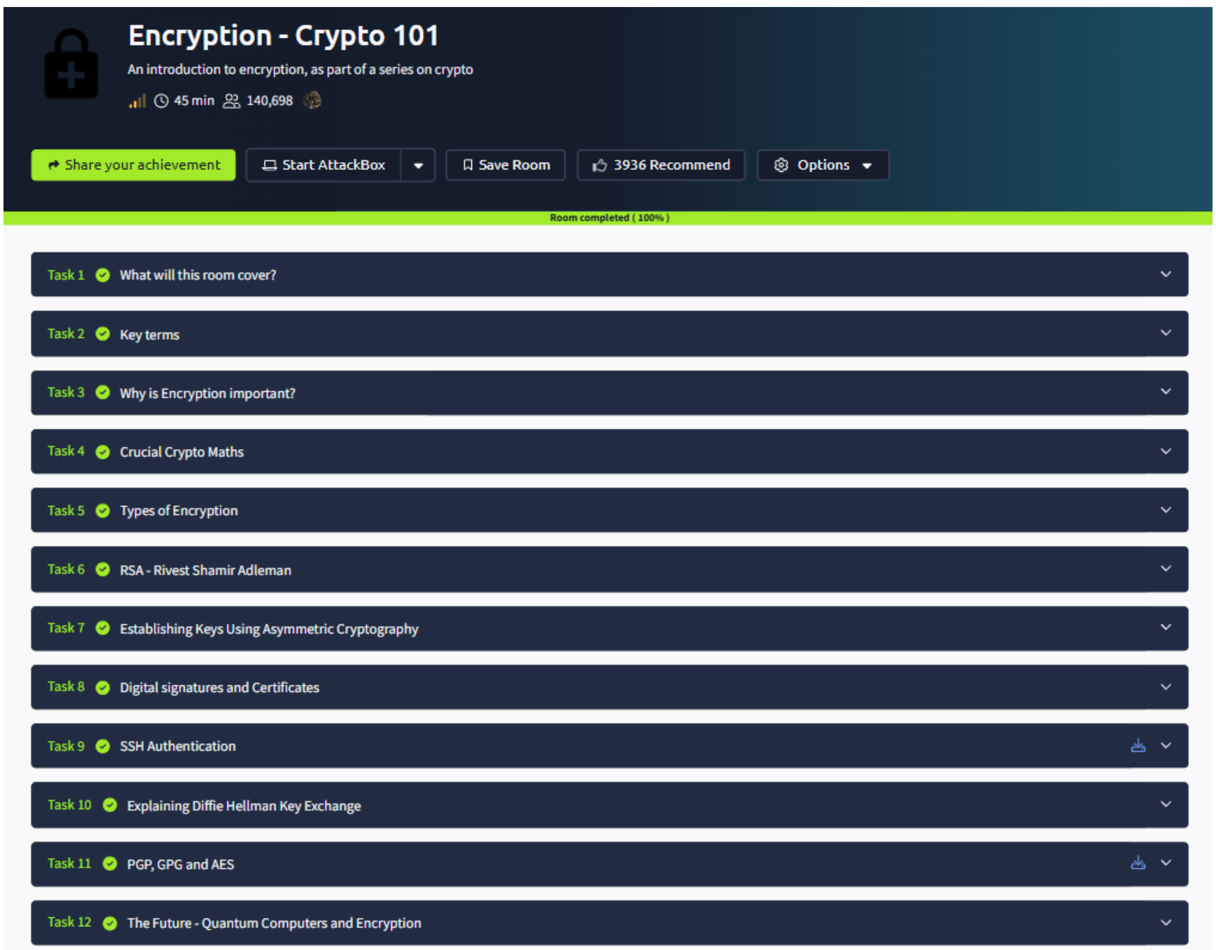
Symmetric	Asymmetric
One key	Two keys

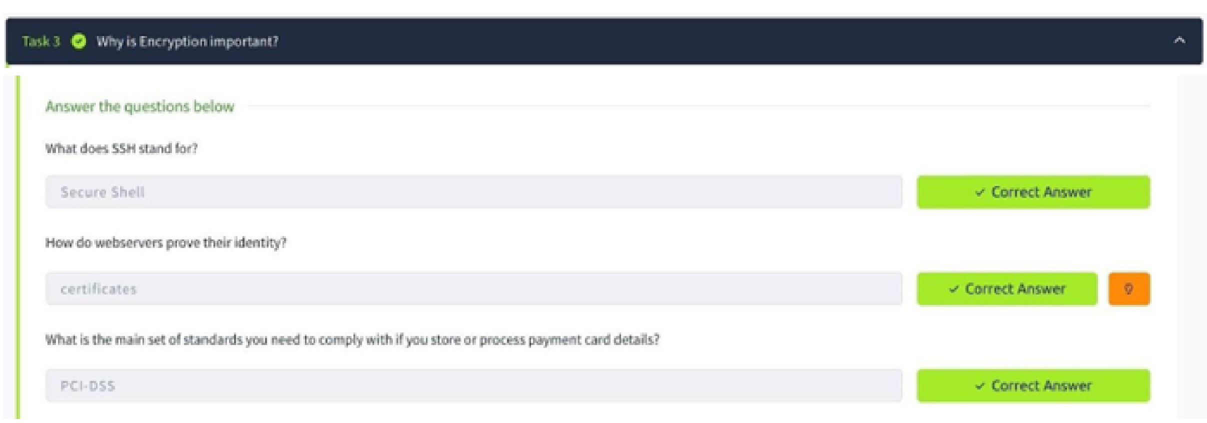
Fast	Slower
Example: AES	Example: RSA

Step 5: Verify Completion

1. Ensure all tasks are marked **Completed**
2. Observe the success message at the end of the room

Output:





Task 4 Crucial Crypto Maths

Answer the questions below

What's 30 % 5?

0 — Correct Answer

What's 25 % 7

4 — Correct Answer

What's 118613842 % 9091

3565 — Correct Answer

Task 5 Types of Encryption

Answer the questions below

Should you trust DES? Yea/Nay

Nay — Correct Answer

What was the result of the attempt to make DES more secure so that it could be used for longer?

Triple DES — Correct Answer

Is it ok to share your public key? Yea/Nay

Yea — Correct Answer

Task 6 RSA - Rivest Shamir Adleman

Answer the questions below

p = 4391, q = 6659. What is n?

29239669 — Correct Answer

I understand enough about RSA to move on, and I know where to look to learn more if I want to.

No answer needed — Correct Answer

Task 7 Establishing Keys Using Asymmetric Cryptography

Answer the questions below

I understand how keys can be established using Public Key (asymmetric) cryptography.

No answer needed — Correct Answer

Task 8 Digital signatures and Certificates

Answer the questions below

What can you use to verify that a file has not been modified and is the authentic file as the author intended?

Digital Signature — Correct Answer

Result:
Thus, the Crypto 101 experiment was successfully completed on the TryHackMe platform.
The experiment helped in understanding basic cryptographic concepts, encryption techniques, encoding methods, and hashing algorithms through hands-on practice.